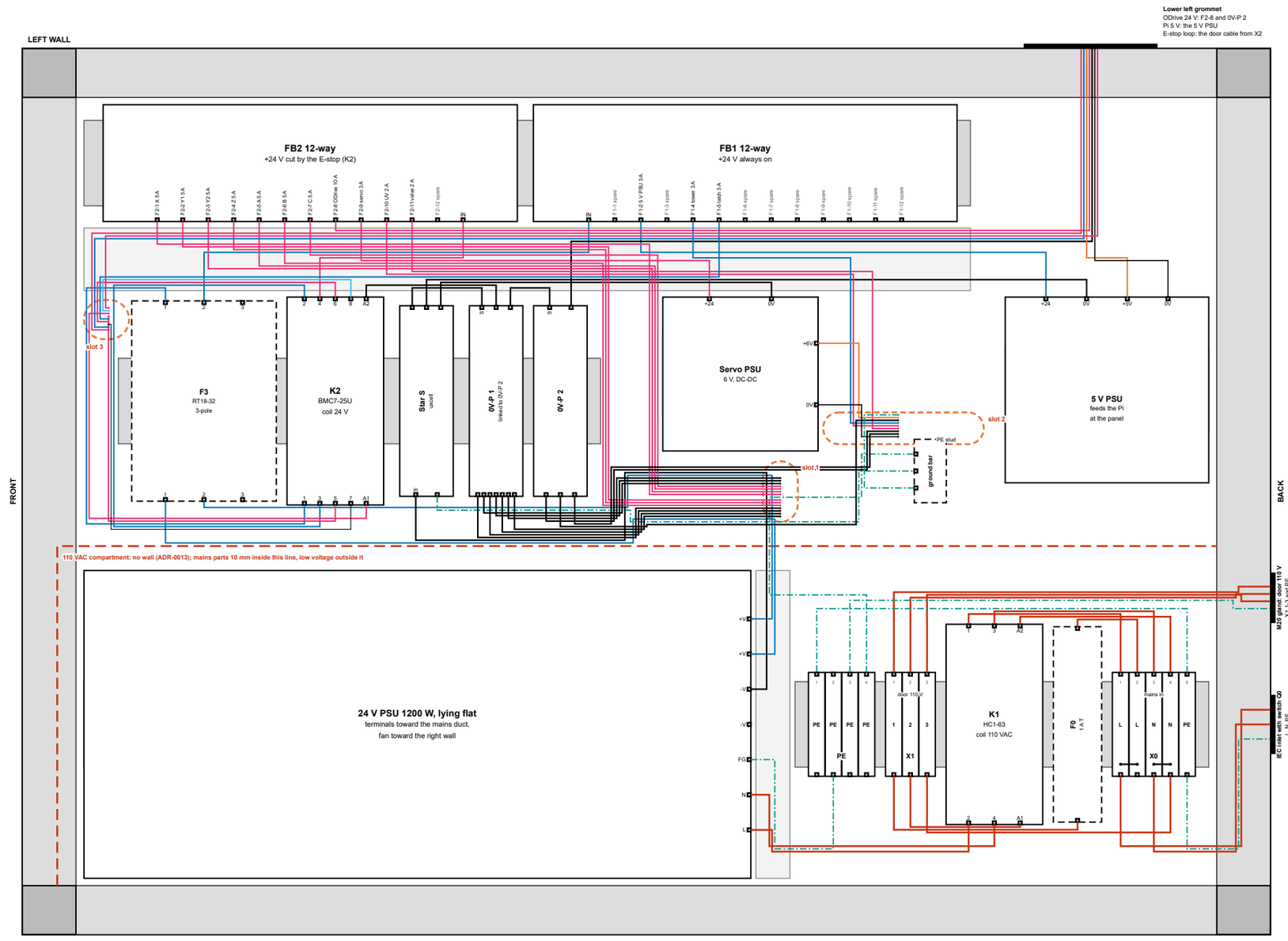
# RMR control panel: lower deck wiring

Lower deck, on the base plate: 110 VAC and 24 V power. Sheet 1's circuits on the parts of sheet 3, every wire from terminal to terminal.




## Legend

- 110 VAC
- +24 V behind K2 (FB2)
- 0V-S signal 0 V
- +6 V servo, +5 V Pi
- signal
- motor cable
- wire duct
- cable slot (dashed: overhead)
- grommet, inlet or gland
- +24 V always on (FB1)
- 0 V power (0V-P, star S)
- PE
- their 0 V
- multi-core signal cable
- DIN rail
- wire ends in a slot
- part (dashed: size assumed)

## Wires between the decks

**Slot 1 (x 267.5-280.5, y 156-178)**

- F2-1 to driver X +V
- F2-2 to driver Y1 +V
- F2-3 to driver Y2 +V
- F2-4 to driver Z +V
- F2-5 to driver A +V
- F2-6 to driver B +V
- F2-7 to driver C +V
- 0V-P 1 to driver X 0V
- 0V-P 1 to driver Y1 0V
- 0V-P 1 to driver Y2 0V
- 0V-P 1 to driver Z 0V
- 0V-P 1 to driver A 0V
- 0V-P 1 to driver B 0V
- 0V-P 1 to driver C 0V

**Slot 2 (x 290-350, y 136-149)**

- F2-10 to SSR-UV +
- F2-11 to SSR-valve +
- F1-4 to the tower SSRs + (jumpered)
- 0V-P 2 to X6 UV-
- 0V-P 2 to X6 V-
- 0V-P 2 to X7 C (tower common)
- servo PSU +6 V to X5 6V
- servo PSU 0 V to X5 0V
- lower ground bar to upper ground bar

**Slot 3 (x 23-36, y 90-106)**

- F1-5 to X2 1 (E-stop latch)
- K2 5 to X2 2b (seal-in, node X)
- K2 6 to X2 3 (node Y)
- K2 A1 to X2 3b (coil, node Y)
- K2 7 to Mega 43 (motor power sense)
- K2 8 to 0V-S
- door cable: X2 1 to S2 NC1, via the lower left grommet
- door cable: NC1 and S3 (node X) to X2 2
- door cable: S3 (node Y) to X2 3

## Notes

Seen from above, front at the left, so the left wall is at the top. Each part sits where sheet 3 puts it; ducts and channels are drawn wider than scale so that every wire shows: measure from sheet 3. Terminals are drawn on the face their wires leave from; read the real order off each part.

Since 2026-10-15 the Pi sits at the operator panel on the 5 V PSU, the Mega runs on the PI's USB and the screen has no supply of its own: F1-1, F1-3, X8 and X2 4-5 are spare (NC2 and the home button are wired at the panel). The ODrive sits in the spin coater's base.

Gauges (issue #205): mains, the PSU to F3 and S, F3 to K2 and FB1, K2's pole jumper, K2 to FB2, S to 0V-P and the 0V-P link: 12 AWG copper. F2-1 to F2-8 with their 0 V returns, and the servo 6 V and Pi 5 V pairs: 14 AWG copper. The F0 rung: 18 AWG UL 1015, 600 V. Everything else: the 20 AWG on hand, copper-clad aluminum (about 3 A), only on fuses of 3 A or less.

PE home runs of the frame, the DIN rails and the TB6600 cases go to the ground bars (sheet 4). 0 V meets PE only at star S, by its one wire to the lower ground bar.

Open: tower common polarity, lid feed, valve coil, servo 6 V rating, J11 logic supply, and the 5 V PSU's current for the Pi 5 (5.1 V, 5 A).

Mains duct: a separator between the 110 VAC wires and the PSU's DC and PE leads. Every 110 VAC terminal finger-safe or covered (ADR-0013). Fuses: F3-1 25 A and F3-2 10 A (10 x 38 mm, DC), F2-1 to F2-7 5 A, F2-8 10 A, F2-9 3 A, F2-10 and F2-11 2 A; F1-2, F1-4 and F1-5 3 A; F0 1 A T.

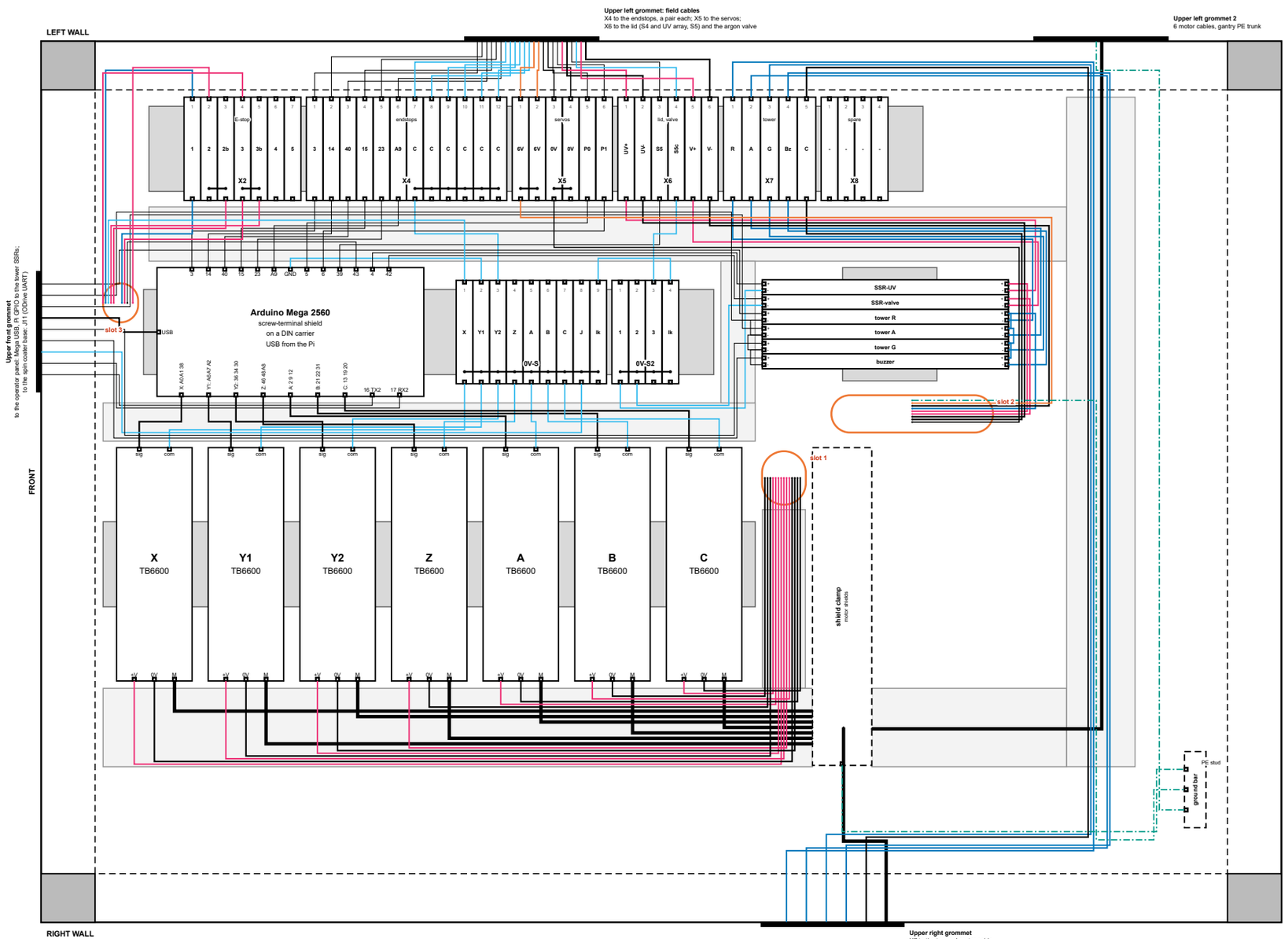
The PSU's terminal order is assumed.

## RMR control panel: lower deck wiring

Sheet 2A (2B: upper deck). Legal 14 x 8.5 in, landscape, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-19 (issue #164), revised 2026-10-22.
Status: design, not yet built or tested.
Circuits: sheet 1. Positions: sheet 3. Frame and PE: sheet 4.

# RMR control panel: upper deck wiring

Upper deck, on the upper plate: drives, logic and field terminals. Sheet 1's circuits on the parts of sheet 3, every wire from terminal to terminal.




## Legend

- 110 VAC
- +24 V behind K2 (FB2)
- 0V-S signal 0 V
- +6 V servo, +5 V Pi
- signal
- motor cable
- wire duct
- cable slot (dashed: overhead)
- grommet, inlet or gland
- +24 V always on (FB1)
- 0 V power (0V-P, star S)
- PE
- their 0 V
- multi-core signal cable
- DIN rail
- wire ends in a slot
- part (dashed: size assumed)

## Wires between the decks

**Slot 1 (x 267.5-280.5, y 156-178)**

- F2-1 to driver X +V
- F2-2 to driver Y1 +V
- F2-3 to driver Y2 +V
- F2-4 to driver Z +V
- F2-5 to driver A +V
- F2-6 to driver B +V
- F2-7 to driver C +V
- 0V-P 1 to driver X 0V
- 0V-P 1 to driver Y1 0V
- 0V-P 1 to driver Y2 0V
- 0V-P 1 to driver Z 0V
- 0V-P 1 to driver A 0V
- 0V-P 1 to driver B 0V
- 0V-P 1 to driver C 0V

**Slot 2 (x 290-350, y 136-149)**

- F2-10 to SSR-UV +
- F2-11 to SSR-valve +
- F1-4 to the tower SSRs + (jumpered)
- 0V-P 2 to X6 UV-
- 0V-P 2 to X6 V-
- 0V-P 2 to X7 C (tower common)
- servo PSU +6 V to X5 6V
- servo PSU 0 V to X5 0V
- lower ground bar to upper ground bar

**Slot 3 (x 23-36, y 90-106)**

- F1-5 to X2 1 (E-stop latch)
- K2 5 to X2 2b (seal-in, node X)
- K2 6 to X2 3 (node Y)
- K2 A1 to X2 3b (coil, node Y)
- K2 7 to Mega 43 (motor power sense)
- K2 8 to 0V-S
- door cable: X2 1 to S2 NC1, via the lower left grommet
- door cable: NC1 and S3 (node X) to X2 2
- door cable: S3 (node Y) to X2 3

## Notes

Seen from above, front at the left, so the left wall is at the top. Each part sits where sheet 3 puts it; ducts and channels are drawn wider than scale so that every wire shows: measure from sheet 3. Terminals are drawn on the face their wires leave from; read the real order off each part.

Since 2026-10-15 the Pi sits at the operator panel on the 5 V PSU, the Mega runs on the PI's USB and the screen has no supply of its own: F1-1, F1-3, X8 and X2 4-5 are spare (NC2 and the home button are wired at the panel). The ODrive sits in the spin coater's base.

Gauges (issue #205): mains, the PSU to F3 and S, F3 to K2 and FB1, K2's pole jumper, K2 to FB2, S to 0V-P and the 0V-P link: 12 AWG copper. F2-1 to F2-8 with their 0 V returns, and the servo 6 V and Pi 5 V pairs: 14 AWG copper. The F0 rung: 18 AWG UL 1015, 600 V. Everything else: the 20 AWG on hand, copper-clad aluminum (about 3 A), only on fuses of 3 A or less.

PE home runs of the frame, the DIN rails and the TB6600 cases go to the ground bars (sheet 4). 0 V meets PE only at star S, by its one wire to the lower ground bar.

Open: tower common polarity, lid feed, valve coil, servo 6 V rating, J11 logic supply, and the 5 V PSU's current for the Pi 5 (5.1 V, 5 A).

Motor cables: shield to the clamp bar; six leave by the upper left grommet and one by the upper right one: pick which on the bench. Each TB6600: PUL-, DIR- and ENA- jumpered to COM at the driver. The tower cable leaves X7 by the upper right grommet, as sheet 3 has it, so it crosses the deck behind every rail; the upper left grommet, beside X7, would save most of that run.

1 k from 5V to 43 and 100 nF from 40 to GND on the Mega's screw shield.

## RMR control panel: upper deck wiring

Sheet 2B (2A: lower deck). Legal 14 x 8.5 in, landscape, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-19 (issue #164), revised 2026-10-22.
Status: design, not yet built or tested.
Circuits: sheet 1. Positions: sheet 3. Frame and PE: sheet 4.

# RMR control panel: names and terminals

Sheet 2C: what each name on sheets 1, 2A and 2B is and where it sits, and what lands on every terminal. The terminal plan is read from the wire lists that draw sheets 2A and 2B.

## Lower deck: 110 VAC and 24 V power

| | | |
|---|---|---|
| **Q0** | switch on the IEC inlet: isolates the mains | back wall |
| **X0** | mains in: L, L, N, N, PE from the inlet | 110 V rail |
| **F0** | fuse holder, 1 A T: feeds the K1 rung | 110 V rail |
| **K1** | 110 VAC contactor HC1-63: mains to the 24 V PSU, coil by S1 | 110 V rail |
| **X1** | door 110 V: out to S1 and H1 on the panel | 110 V rail |
| **PE** | 4 ground blocks: inlet PE, PSU FG, door plate, lower bar | 110 V rail |
| **PSU** | 24 V 1200 W supply, lying flat, fan to the right wall | 110 VAC area |
| **FB1** | 12-way fuse module: +24 V always on | fuse rail |
| **FB2** | 12-way fuse module: +24 V cut by the E-stop (behind K2) | fuse rail |
| **F3** | 3-pole fuse holder RT18-32, 10 x 38 mm: F3-1 25 A from PSU V+ to K2 (FB2), F3-2 10 A to FB1, F3-3 spare | 24 V rail |
| **K2** | 24 V contactor BMC7-25U: 1-2 and 3-4 in series feed FB2, 5-6 seal in, 7-8 tell Mega pin 43 that motor power is on | 24 V rail |
| **star S** | the 0 V star at the PSU's -V; the only 0 V to PE bond | 24 V rail |
| **0V-P 1, 2** | 0 V power buses, linked: drivers, ODrive, K2 coil, loads | 24 V rail |
| **servo PSU** | 6 V DC-DC for the gripper and lid servos | beside 24 V rail |
| **5 V PSU** | feeds the Pi at the operator panel | 24 V row, back |
| **ground bar** | lower PE bar on the base plate's PE stud | behind servo PSU |

## Upper deck: drives, logic, field terminals

| | | |
|---|---|---|
| **X2** | E-stop latch: the door cable from S2 and S3 | field rail |
| **X4** | endstops: a signal and a common for each axis | field rail |
| **X5** | servos: 6 V, 0 V and signal for the gripper and lid | field rail |
| **X6** | lid (UV array, switch S5) and the argon valve | field rail |
| **X7** | status tower: red, amber, green, buzzer, common | field rail |
| **X8** | spare (was the screen supply and the home button) | field rail |
| **Mega** | Arduino Mega 2560 on a screw shield; runs on the Pi's USB | logic rail |
| **0V-S** | signal 0 V bus, 9 linked: driver COMs, Mega GND, endstops | logic rail |
| **0V-S2** | second 0V-S group: SSR input returns, S5 return | logic rail |
| **SSR-UV** | DC solid-state relay: Mega pin 4 switches the UV array | SSR rail |
| **SSR-valve** | DC solid-state relay: Mega pin 42 switches the argon valve | SSR rail |
| **tower R A G** | DC SSRs: Pi GPIO 5, 6, 13 switch the red, amber, green lamps | SSR rail |
| **buzzer** | DC SSR: Pi GPIO 26 switches the tower's buzzer | SSR rail |
| **X ... C** | TB6600 drivers: X, Y1, Y2 (follows Y), Z, A (filter feed), B (syringe height), C (syringe) | drivers rail |
| **shield clamp** | grips the motor cables' shields, bonded to PE | past the drivers |
| **ground bar** | upper PE bar on the upper plate's PE stud | behind back run |

## Outside the box (sheet 1)

| | | |
|---|---|---|
| **S1, H1** | panel knob that switches K1 on; H1 its lamp, lit while K1 is in | operator panel |
| **S2** | E-stop mushroom: NC1 drops K2, NC2 tells the Pi (GPIO 20) | operator panel |
| **S3** | Reset push button (NO): pulls K2 in after an E-stop | operator panel |
| **S4** | lid switch in series with the UV array: closed = lid shut | lid |
| **S5** | lid switch to Mega pin 39 (UV_LID_INTERLOCK, off for now) | lid |
| **D1** | 1N4007 across the argon valve's coil, band to + | at the valve |
| **Pi** | Raspberry Pi 5 with the touch screen, runs rmr_bridge.py | operator panel |
| **ODrive** | ODrive S1 spin coater drive; J11 is its UART | spin coater base |
| **X, Y (nodes)** | E-stop latch nodes: X after NC1, Y after S3 (sheet 1, rung B) | |

## Terminal plan

| No. | label | in the box (by) | field side (by) |
|---|---|---|---|
| **X2 (upper deck)** | | | |
| **1** | 1 | door cable, LL (slot 3); F1-5 (slot 3) | +24 V to S2 NC1 |
| **2** | 2 | door cable, LL (slot 3) | NC1 out, S3 in (node X) |
| **3** | 2b | K2 5 (slot 3) | - |
| **4** | 3 | door cable, LL (slot 3); K2 6 (slot 3) | S3 out (node Y) |
| **5** | 3b | K2 A1 (slot 3) | - |
| **6** | 4 | - | spare: NC2 is wired at the panel |
| **7** | 5 | - | spare |
| **X4 (upper deck)** | | | |
| **1** | 3 | Mega 3 | endstop X signal (UL) |
| **2** | 14 | Mega 14 | endstop Y signal (UL) |
| **3** | 40 | Mega 40 | endstop Z signal (UL) |
| **4** | 15 | Mega 15 | endstop A signal (UL) |
| **5** | 23 | Mega 23 | endstop B signal (UL) |
| **6** | A9 | Mega A9 | endstop C signal (UL) |
| **7** | C | 0V-S | endstop X common (UL) |
| **8** | C | bridged to 7 | endstop Y common (UL) |
| **9** | C | bridged to 7 | endstop Z common (UL) |
| **10** | C | bridged to 7 | endstop A common (UL) |
| **11** | C | bridged to 7 | endstop B common (UL) |
| **12** | C | bridged to 7 | endstop C common (UL) |
| **X5 (upper deck)** | | | |
| **1** | 6V | servo PSU OUT+ (slot 2) | gripper servo +6 V (UL) |
| **2** | 6V | bridged to 1 | lid servo +6 V (UL) |
| **3** | 0V | servo PSU OUT- (slot 2) | gripper servo 0 V (UL) |
| **4** | 0V | bridged to 3 | lid servo 0 V (UL) |
| **5** | P0 | Mega 5 | gripper servo signal (UL) |
| **6** | P1 | Mega 6 | lid servo signal (UL) |
| **X6 (upper deck)** | | | |
| **1** | UV+ | SSR-UV out - | S4, then the UV array + (UL) |
| **2** | UV- | 0V-P 2 (slot 2) | UV array - (UL) |
| **3** | S5 | Mega 39 | lid switch S5 (UL) |
| **4** | S5c | 0V-S2 | S5's other side (UL) |
| **5** | V+ | SSR-valve out - | argon valve + (D1 across) (UL) |
| **6** | V- | 0V-P 2 (slot 2) | argon valve - (UL) |
| **X7 (upper deck)** | | | |
| **1** | R | tower R out - | tower red (UR) |
| **2** | A | tower A out - | tower amber (UR) |
| **3** | G | tower G out - | tower green (UR) |
| **4** | Bz | buzzer out - | tower buzzer (UR) |
| **5** | C | 0V-P 2 (slot 2) | tower common (UR) |
| **X8 (upper deck)** | | | |
| **1** | - | - | - |
| **2** | - | - | - |
| **3** | - | - | - |
| **4** | - | - | - |
| **0V-S (upper deck)** | | | |
| **1** | X | K2 8 (slot 3); driver X COM | - |
| **2** | Y1 | Mega GND; driver Y1 COM | - |
| **3** | Y2 | X4 7; driver Y2 COM | - |
| **4** | Z | driver Z COM | - |
| **5** | A | driver A COM | - |
| **6** | B | driver B COM | - |
| **7** | C | driver C COM | - |
| **8** | J | bridged to 1 | ODrive J11 GND (UF) |
| **9** | lk | 0V-S2 | - |
| **0V-S2 (upper deck)** | | | |
| **1** | 1 | SSR-UV in - | - |
| **2** | 2 | SSR-valve in - | - |
| **3** | 3 | X6 4 | - |
| **4** | lk | 0V-S | - |

## X0 (lower deck)

| | | | |
|---|---|---|---|
| **1** | L | K1 1 | L from the inlet (inlet) |
| **2** | L | F0 in | - |
| **3** | N | K1 3 | N from the inlet (inlet) |
| **4** | N | K1 A2; X1 3 | - |
| **5** | PE | PE 1 | PE from the inlet (inlet) |

## X1 (lower deck)

| | | | |
|---|---|---|---|
| **1** | 1 | F0 out | to S1 (gland) |
| **2** | 2 | K1 A1 | from S1: K1 coil, H1 (gland) |
| **3** | 3 | X0 4 | N to H1 (gland) |

## PE (lower deck)

| | | | |
|---|---|---|---|
| **1** | PE | X0 5 | - |
| **2** | PE | PSU FG | - |
| **3** | PE | - | door plate PE (gland) |
| **4** | PE | ground bar | - |

## Buses on the lower deck

| | |
|---|---|
| **star S** | 0V-P 1; PSU -V; 5 V PSU IN-; ground bar; servo PSU IN- |
| **0V-P 1** | driver A 0V (slot 1); driver B 0V (slot 1); driver C 0V (slot 1); star S; K2 A2; 0V-P 2; driver X 0V (slot 1); driver Y1 0V (slot 1); driver Y2 0V (slot 1); driver Z 0V (slot 1) |
| **0V-P 2** | 0V-P 1; ODrive DC- (LL); X7 5 (slot 2); X6 2 (slot 2); X6 6 (slot 2) |

## Fuses

| | |
|---|---|
| **FB1** | F1-1 spare, F1-2 5 V PSU (3 A), F1-3 spare, F1-4 tower (3 A), F1-5 latch (3 A), F1-6 spare, F1-7 spare, F1-8 spare, F1-9 spare, F1-10 spare, F1-11 spare, F1-12 spare |
| **FB2** | F2-1 X (5 A), F2-2 Y1 (5 A), F2-3 Y2 (5 A), F2-4 Z (5 A), F2-5 A (5 A), F2-6 B (5 A), F2-7 C (5 A), F2-8 ODrive (10 A), F2-9 servo (3 A), F2-10 UV (2 A), F2-11 valve (2 A), F2-12 spare |
| **F0** | 1 A T, 5 x 20 mm: the K1 rung (S1, K1 coil, H1) |
| **F3** | 10 x 38 mm, DC rated: F3-1 25 A, PSU V+ to K2 (FB2); F3-2 10 A, PSU V+ to FB1; F3-3 spare |

## Cable entries and slots

| | |
|---|---|
| **UL** | upper left grommet: field cables X4, X5, X6 |
| **ULM** | upper left grommet 2: 6 motor cables, gantry PE |
| **UF** | upper front grommet: Mega USB, J11, Pi GPIO |
| **UR** | upper right grommet: X7 tower, 1 motor cable |
| **LL** | lower left grommet: ODrive 24 V, Pi 5 V, door cable |
| **inlet** | IEC inlet with switch Q0, back panel |
| **gland** | M20 gland, back panel: door 110 V cable |
| **slot 1** | upper plate: driver +V and 0 V |
| **slot 2** | upper plate: SSR feeds, returns, servo 6 V, PE link |
| **slot 3** | upper plate: E-stop latch, pin 43, door cable |

## Abbreviations

| | |
|---|---|
| **PE** | protective earth |
| **FG** | the PSU's frame ground (earth) terminal |
| **NC, NO** | normally closed, normally open contact |
| **SSR** | solid-state relay |
| **COM** | common |
| **PUL, DIR, ENA** | a TB6600's step, direction, enable inputs |
| **F1-n, F2-n, F3-n** | fuse n of FB1, of FB2, of F3 |
| **0V-P, 0V-S** | 0 V power, 0 V signal |
| **GPIO** | a Pi input/output pin (BCM numbers) |
| **J11** | the ODrive's isolated UART connector |
| **RX2, TX2** | the Mega's Serial2 pins (17, 16) |
| **USLKG 5** | DIN ground terminal block |
| **UK 2.5 B** | DIN feed-through terminal block |
| **TS35** | 35 mm top-hat DIN rail |
| **SJOOW** | oil-resistant portable cord, 300 V |
| **AWG** | American wire gauge |

## RMR control panel: names and terminals

Sheet 2C (2A, 2B: the decks; 2D: the panel). Legal 14 x 8.5 in, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-20 (issue #171).
Status: design, not yet built or tested.
Circuits: sheet 1. Positions: sheet 3. Wiring: sheets 2A, 2B.

# RMR control panel: operator panel

Sheet 2D: how the operator panel might look from the front, at 1:1.5 with assumed sizes, and what each control wires to. The dashed outline is the panel as it may grow.




10.1 in HDMI touch screen, Pi 5 behind it (Touch UI, Jog tab)

## Behind the panel

| control | contact | lands on | cable (by) |
|---|---|---|---|
| **S1 power knob** | NO, 110 V | X1 1 in, X1 2 out | 110 V door cable (M20 gland) |
| **H1 lamp, in S1** | 110 V | X1 2 and X1 3 (N) | 110 V door cable |
| **panel PE** | | PE 3 | 110 V door cable |
| **S2 E-stop** | NC1 | X2 1 (+24 V), X2 2 (node X) | E-stop loop (LL, slot 3) |
| **S2 E-stop** | NC2 | Pi GPIO 20 and GND, 1 k to 3V3 | at the panel |
| **S3 Reset** | NO | X2 2 (node X) to X2 3 (node Y) | E-stop loop (LL, slot 3) |
| **touch screen** | HDMI, USB | Pi 5 | at the panel |
| **Pi 5** | 5 V in | 5 V PSU (F1-2) | 5 V feed (LL) |
| **Pi 5** | USB | Mega | USB cable (UF) |
| **Pi 5** | GPIO 5, 6, 13, 26 | tower SSRs (GND to their inputs -) | tower inputs (UF) |

## RMR control panel: operator panel

Sheet 2D (2A, 2B: the decks; 2C: names). Legal 14 x 8.5 in, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-20 (issue #171).
Status: concept; housing and sizes not designed yet.
Circuits: sheet 1. Terminals: sheet 2C.